

ImageBind Guided Progressive Transformation Network for Alignment-free RGBT Video Object Detection

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Abstract

Existing RGB and Thermal video object detection (RGBT VOD) models typically rely on cross-modal aligned data, while alignment process is usually manual and thus time-consuming and labor-intensive. Cross-modal alignment methods are often adopted to mitigate the spatial differences of unaligned data, but would suffer from poor generalization due to the intrinsic modality gap between RGB and thermal modalities in challenging scenarios. To this end, this paper presents a novel ImageBind guided Progressive Transformation Network called IPTNet, which leverages the generalization ability of pretrained multimodal foundation model to adapt diverse challenging cross-modal alignment scenarios, for alignment-free RGBT VOD. In particular, we design an aligner with a corresponding alignment consistency all built upon the pretrained multimodal foundation model. The aligner adopts a progressive two-stage alignment strategy, consisting of a linear affine transformation module for coarse locating and a warping-based affine module for fine-grained refinement. Each stage is guided by a dedicated alignment consistency loss to ensure progressively constrained and accurate alignment in the multimodal feature space. In addition, to fully exploit the benefits of multimodal fusion under severe modality receptive field discrepancy, we design a Thermal-to-RGB fusion module to enable high-fidelity feature integration in a specific region of thermal modality. Our proposed IPTNet shows superior performance compared with existing RGBT VOD methods on the challenging UVT-VOD2024, VT-VOD50, CVC-14 and LLVIP benchmark datasets. Especially on the UVT-VOD2024 dataset, our method surpasses its benchmark by approximately 13%.

Keywords: RGBT Video Object Detection, Multi-modal Fusion, Alignment-free Detection, ImageBind Foundation Model

1 Introduction

RGB-based object detection methods face inherent limitations in challenging environments such as low-light conditions, cluttered backgrounds, and adverse weather. These constraints often degrade the visibility of objects in RGB images and influence object

¹ detection performance, highlighting the need for mul- 7
timodal object detection approaches. The fusion of 8
² RGB and thermal (RGBT) modalities has emerged 9
³ as a promising solution, leveraging thermal image’s 10
⁴ robustness to illumination variations. Nevertheless, 11
⁵ the original image pairs captured by RGBT sensors are 12
⁶ not aligned, even by the most advanced one. Existing 13
RGBT detection frameworks predominantly rely on 14

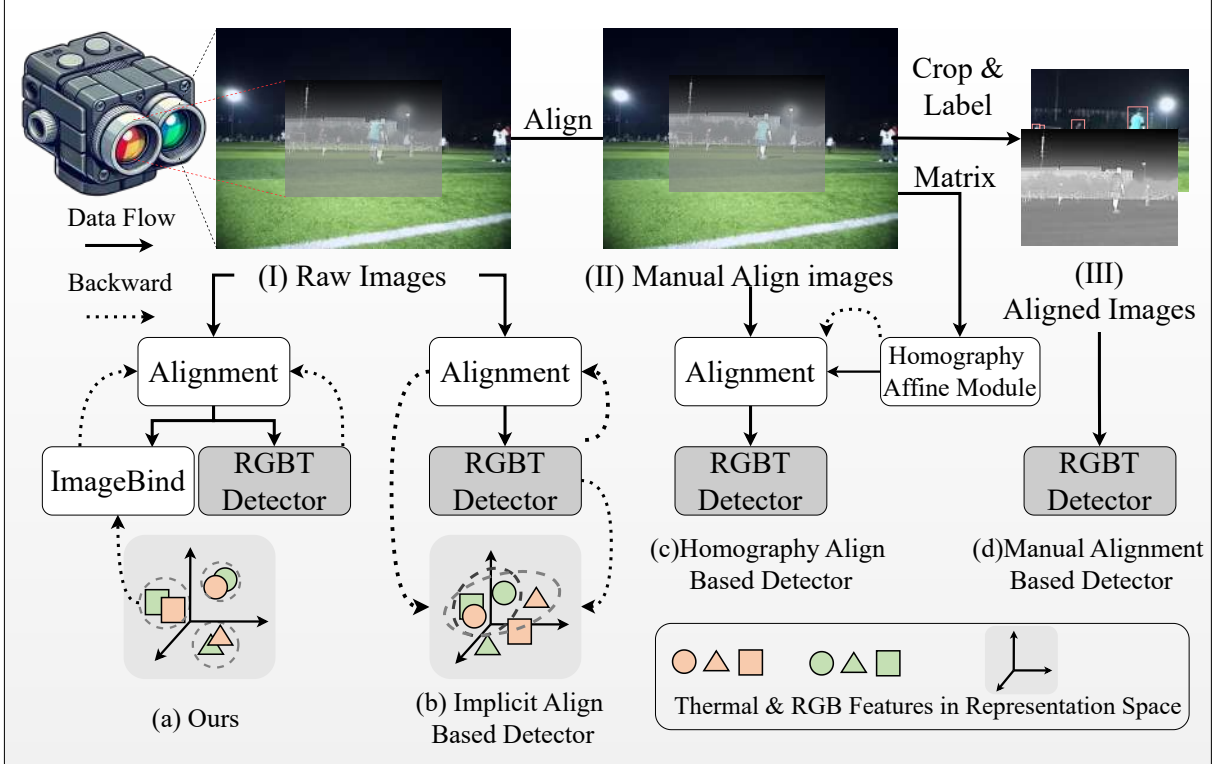


Fig. 1: RGBT object detection process with different alignment strategy. (d) is the most common detection process based on manual aligned RGBT dataset, costing too much labor. (c) tries to learn the alignment with extra homography groundtruth, still suffers from the high cost of human intervention; (b) uses the original image pairs to achieve alignment under self-supervision, gradually learning the optimal solution, but neglects the gap between modalities; (a) is our method, which uses pre-trained ImageBind model to achieve cross-modal alignment, avoiding the cross-modal difference.

manually aligned multimodal data [48, 27, 34] to train models, as illustrated in Fig. 1(d). However, the manual alignment process needs substantial annotations while failing to address some misalignments caused by sensor-specific disparities in focal length, view-point, etc. That is why cropping is carried out at the end of alignment, with the aim of avoiding these complex challenges caused by original RGBT data, which leads to poor performance in real applications.

Current RGBT methodologies of addressing misalignment fall into two mainstream paradigms, both with fundamental limitations. Homography-based approaches [33, 16] attempt cross-modal alignment through affine transformation parameter estimation, as shown in Fig. 1(c). However, they require precise homography matrix annotations for supervised learning, incurring costs comparable to manual alignment. More critically, these methods fail to address complex nonlinear misalignments caused by two sensors'

parallax, as homography matrices can't characterize perspective deformations, while this problem can be avoided by cropping in Fig. 1(III). This essentially shifts the annotation demand from pixel alignment to matrix parameter labeling, without achieving genuine unsupervised or self-supervised alignment.

Self-supervised or unsupervised methods [6, 47] achieve cross-modal alignment by projecting features from both modalities into a shared latent space. These methods often incorporate additional loss and delicately designed modules at various level in the network, aiming to enhance alignment effect for downstream tasks. However, this alignment remains essentially not robust, which is learned from synthetic artificial constraints rather than knowledge in real-world geometric or semantic correspondences. Such an operation often results in poor generalization ability of these models, making unsatisfactory performances when faced with more complex and diverse

misalignment situations. Ultimately, the model learns an implicit latent space that is optimized to the best extent possible under a given loss, as illustrated in Fig. 1(b), struggling to bridge the inherent differences between modalities. In fact, cross-modal alignment modules essentially learn from scratch the ability to bridge the inherent gap between modalities, which becomes particularly challenging in the absence of large-scale ground-truth supervision.

To address these issues, we propose a novel ImageBind guided Progressive Transformation Network called IPTNet, a self-supervised alignment method based on the pretrained model-driven mechanism to mitigate the effects of sensors' differences and dynamic scene variations, ensuring the efficient and accurate object detection in complex environments. We think that an effective cross-modal alignment that can break the gap between modalities should transition from high-level semantic consistency to low-level spatial alignment, where a successful alignment reflects a comprehensive understanding of global context. We have noticed that the pre-trained model ImageBind [20] possesses the remarkable capability in establishing effective connections between different modalities within a shared semantic space. ImageBind focuses on image-centric training, aligning the modalities in pairs. As a result, it includes image-level alignment knowledge and semantic extension ability from images to other modalities. These priors are crucial for achieving the effective alignment in challenging scenarios without influencing semantic integrity.

In IPTNet, a Two-stage Aligner with a corresponding alignment consistency all built upon the pretrained multimodal foundation model is designed to achieve progressive alignment. In particular, the complete RGB image and the constrained thermal image are first passed through the frozen encoders of the corresponding modalities in the ImageBind model to obtain their embeddings. The Linear Transformation module then predicts the global offset between the modalities based on the two embeddings, generating a linear transformation matrix used to fine-tune the size of the thermal image and locate its position relative to the RGB image. Next, after the linear alignment, the thermal and RGB images undergo a Warp Transformation module which progressively processes dense matching to calculate the cross-modal fine-grained pixel differences, resulting in a non-linear deformation field. This field is used to significantly adjust the thermal image from the previous stage, for addressing

the challenging alignment. To manage the instability of aligner, the final thermal image is re-constrained by the RGB one in the ImageBind model's feature space. After the alignment, the thermal and RGB features are processed through Thermal-to-RGB fusion (T2RF) module to maximize the use of information from thermal modality without being affected by low receptive field fusion of two modalities. Finally, we extend the encoder of DINO [58], which is a deformable DETR-like detector, to build temporal connection to achieve information complementation, for obtaining the predicted result.

IPTNet is formulated as an RGBT video object detection framework because it is first designed for native unaligned RGBT video streams. For a video stream, reliable frame-level cross-modal processing is still the foundation of the whole detection pipeline. Therefore, IPTNet keeps the alignment and fusion modules at the frame level, making them naturally applicable to image-based RGBT detection as well. Overall, our goal is to address the alignment-free RGBT setting in both image pairs and video streams, while maintaining a balanced design among cross-modal alignment, feature fusion, and temporal aggregation. The main contributions of the paper are summarized as follows:

- We propose the IPTNet for alignment-free RGBT VOD, which achieves cross-modal alignment in an end-to-end manner with raw RGBT image pairs, without the geometric alignment supervision, effectively addressing complex unalignment challenges like resolution gap and viewpoint variation.
- We design a two-stage aligner whose progressive alignment strategy enables more stable and controllable optimization by leveraging ImageBind's semantic alignment priors, demonstrating its robustness to the modality gap while precisely capturing the differences between modality-specific feature spaces.
- We introduce the T2RF module, which achieves cross-modal local foreground feature integration. This approach leverages the priors of the thermal image in a way suitable for resolution gap between the RGB and the thermal images.
- We conduct comprehensive experiments to verify the effectiveness of the proposed method on four benchmark datasets, namely UVT-VOD2024, CVC-14, VT-VOD50 and LLVIP. Experimental results demonstrate the superiority of the proposed IPTNet.

2 Related Work

2.1 Video Object Detection

The field of Video Object Detection has made significant strides. Many state-of-the-art methods aim to aggregate temporal information from multiple frames to enhance the features of each frame. Temporal fusion aims to enhance detection accuracy. FGFA [70] and THP [68] use the optical flow to align and aggregate frames' features. However, they face the challenges, such as requiring large amounts of flow data and having difficulty in integrating flow extraction into the detection process. Moreover, optical flow struggles to capture long-range dependencies and comes with high computational costs. Techniques like SELSA [50], RDN [15], and MEGA [11] use attention mechanisms and deformable convolutions to aggregate features across frames. These approaches model spatial-temporal contexts effectively, focusing on both local and global information. However, they often involve complex architectures and post-processing methods like Seq-NMS [25], making them sophisticated and resource-heavy.

A recurring issue in VOD is the degradation of detection quality in certain frames, which can be mitigated by using temporal information from other frames. Some methods leveraging optical flow or attention mechanisms aim to strengthen keyframe's features through reference frames, but optical flow-based methods face limitations in computational cost and long-range feature aggregation. More recent methods like TransVOD [66] leverage Transformer-based architectures to improve detection efficiency and accuracy. The success of TransVOD is largely attributed to the powerful capabilities of Deformable DETR [69]. Inspired by similar architecture like [32, 36], our method adopts deformable attention for simply temporal modeling, enabling an efficient and lightweight interaction across adjacent frames.

2.2 Weakly Alignment-free RGBT Object Detection

Recent RGBT object detection work, attempts to address sensors' imaging limitation by incorporating both RGB and thermal data. MS-DETR [53] addresses alignment and imbalance issues between RGB and thermal modalities through a loosely coupled fusion strategy and an instance-aware modality-balanced optimization. DPDETR [23] introduces a

decoupled position detection framework to mitigate the influence caused by misaligned features in RGB-thermal object detection. It employs a decoupled position multispectral cross-attention module for precise feature fusion and a query-decoupled decoder to optimize object localization. DAMSDet [24] dynamically selects salient features through competitive query selection and adaptively fuses multi-semantic features using a deformable cross-attention module, for the challenging RGB-thermal object detection task. However, these methods fail to provide a clear and intuitive view of cross-modal alignment, lacking rigorous interpretability in modeling the alignment process across modalities.

2.3 Downstream Tasks Assisted By ImageBind

ImageBind [20] is a large-scale multimodal model which employs contrastive learning to capture relationships between different modalities through a shared embedding space, including images, text, audio, IMU, depth, and thermal data. This enables comparison and analysis of different types of data within a unified feature space where other modalities' representations align with the representation of image. In Seeing and Hearing [52], ImageBind is used as a classifier guidance that enables training-free audio-to-image generation by measuring cross-modal alignment during the diffusion denoising process, guiding latent updates via gradient feedback. MoCA [1] exploits ImageBind's feature space to devise a pixel-level matching strategy, enabling unsupervised segmentation without any annotations. OV-AVEL [64] freezes ImageBind as a unified semantic space and proposes a two-branch paradigm that enables training-free open-vocabulary event localization. These methods significantly demonstrate that ImageBind's joint embedding space is inherently well-suited for zero-shot and open-vocabulary scenarios.

3 Method

This section presents details of our model comprehensively. Firstly, we introduce the Cross-modal Emergent Capability of ImageBind, based on which, ImageBind can achieve cross-modal spatial alignment. Then we begin with a brief introduction to the IPTNet framework, followed by a detailed explanation of its architecture. This includes a Two-stage

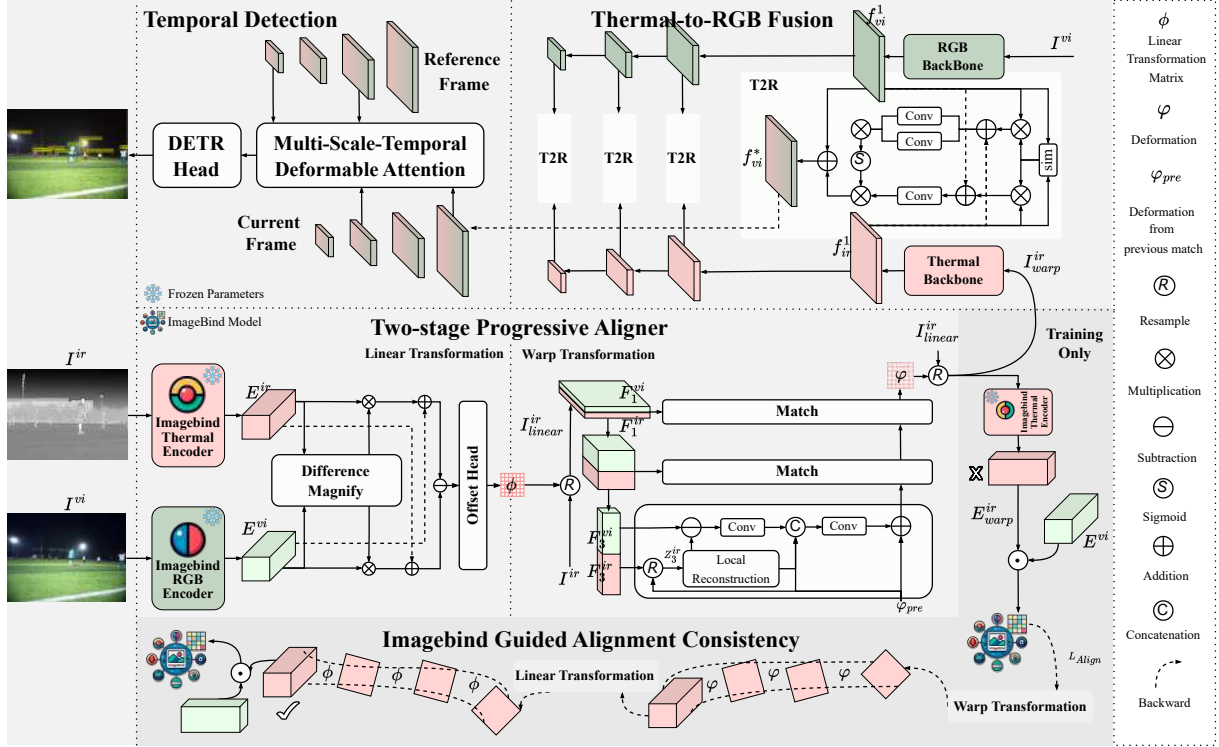


Fig. 2: The proposed IPTNet architecture employs two embeddings from the frozen encoders of ImageBind to predict a linear transformation matrix. This matrix is applied to resample the raw thermal image, enabling cross-modal localization from thermal image to RGB image. Subsequently, dense matching predicts the deformation field between the localized thermal image and the raw RGB image, followed by another resampling operation using this deformation. Through this explicit alignment process, the T2RF module effectively integrates both modalities while accommodating their resolution differences.

progressive Aligner, an ImageBind Guided Alignment Consistency, and a T2RF module.

3.1 Cross-modal Emergent Capability of ImageBind

ImageBind [20] is trained on a massive collection of pairwise aligned multimodal data, aligning them toward the RGB domain through contrastive learning. Although ImageBind is trained with the contrastive learning scheme, its training data is not a single dataset with paired multiple modalities. Rather, it aggregates a large collection of publicly available modality-paired datasets mined from the internet, covering diverse task domains (e.g., image-text retrieval, RGB-depth detection, and RGB-thermal downstream datasets). Importantly, many RGB-Thermal and RGB-Depth data are captured by the calibrated multi-sensor camera and are manually registered. Specifically, multimodal pairs

like RGB-Text and RGB-Audio contribute to global semantic alignment, while pairs such as RGB-thermal, RGB-Depth, and RGB-IMU (Inertial Measurement Unit) that carry geometric cues inherently offer priors for spatial alignment. Undoubtedly, the embedding obtained from a single RGB image via ImageBind contains information from six different modalities. As a result, it can directly function as text in a text-image diffusion model for inference, or directly act as audio in an audio-image diffusion model for reasoning. Building upon this capability, we quantitatively assess the quality of cross-modal spatial alignment in RGBT image pairs by leveraging the consistency captured in these unified embeddings.

3.2 Overview

As shown in Fig. 2, the proposed model is composed of four parts: Two-stage Progressive Aligner,

ImageBind Guided Alignment Consistency, Thermal-to-RGB Fusion, Temporal Detection. Initially, the complete RGB image $I^{vi} \in \mathbb{R}^{H \times W \times 3}$ and the thermal image $I^{ir} \in \mathbb{R}^{H \times W \times 1}$ are passed through frozen modality-specific encoders of ImageBind to extract their respective embeddings $E^m \in \mathbb{R}^{L \times C}, m \in \{vi, ir\}$.

Crucially, each embedding encapsulates cross-modal semantic features within the shared latent space, enabling modality-aware representation learning. Therefore, ImageBind always remains as a permanent component inside Two-stage Progressive Aligner, serving as an indispensable part for obtaining the accurate alignment prior as shown in the overall framework.

The subsequent Linear Transformation module takes embeddings E^m of both modalities as input to predict a transformation matrix $\phi_{ir \rightarrow vi} \in \mathbb{R}^{H \times W \times 2}$ that fine-tunes the size of the thermal image and adjusts its position relative to the RGB image as I_{linear}^{ir} , ensuring a more precise global alignment.

After achieving this linear alignment, we further refine the alignment by performing a Warp Transformation where actually a dense matching is processed $DM_{ir \rightarrow vi}$ between the two modalities. This process calculates the pixel-level cross-modal difference, resulting in a non-linear deformation field $\varphi_{ir \rightarrow vi} \in \mathbb{R}^{H \times W \times 2}$, which is used for a non-linear transformation to adjust the previous thermal image I_{linear}^{ir} to I_{warp}^{ir} , compensating for complex and local misalignments that cannot be resolved by linear transformations alone.

To ensure the stability in handling the progressive alignment process, the final aligned thermal embedding E_{warp}^{ir} from I_{warp}^{ir} is re-constrained with the original RGB embedding E^{vi} within ImageBind’s feature space. This helps mitigate potential instability from non-linear adjustments in each layer and strengthens the overall alignment process. As for this ImageBind Guided Alignment Consistency part, the encoders are for training-only.

Following the cross-modal alignment, we implement the multimodal fusion through the proposed T2RF module. The module operates by: (1) computing cross-modal similarity to selectively enhance RGB features at the region-specific location, and (2) employing a novel homologous attention mechanism where thermal features characterized by their constrained receptive fields simultaneously serve as both queries and keys. This dual-role design enables the

effective capture of thermal-specific patterns, while compensating for inherent limitations in receptive field size or input quality degradation. At each backbone scale level, the fused features subsequently integrate temporal information from reference frames through DINO’s [58] deformable encoder.

3.3 Two-Stage Progressive Aligner

The misalignment presented in the real application poses a critical challenge, which requires us to address the global correspondence of the thermal image relative to the RGB image firstly. This process involves aligning large elements, including background structures and prominent objects, to ensure geometric consistency between thermal and RGB images in terms of vertical positioning and relative scale. By doing so, we achieve a preliminary weak alignment between the thermal image and the RGB image, allowing for basic spatial correspondence. This initial stage of cross-modal localization essentially reflects the manual inspection commonly used in traditional alignment annotation, where coarse matching is performed by visual observation.

After achieving the cross-modal location as mentioned above, the thermal image has undergone rough positioning and global size adjustment. At this point, the misalignment caused by the different vertical positions of the RGB and thermal sensors has been resolved. Especially, background and large objects in the foreground are already aligned, but a horizontal view offset still exists. Hence, a finer adjustment should be conducted to adapt to the difference from two camera sensors’ perspective. In contrast to traditional annotations, our second stage breaks away from the reliance on large-scale cropping of image peripheries to avoid parallax. Instead, we directly model and compensate for the viewpoint differences, allowing for more accurate and complete alignment.

To address this issue, we design a two-stage progressive aligner that simulates the traditional alignment process. The first stage employs a Linear Transformation Module to perform cross-modal localization, enabling a coarse alignment between the thermal and the RGB images. This stage approximates manual alignment by establishing global correspondences in position and scale. The second stage introduces a Warp Transformation Module, which progressively refines the alignment through multi-level matching, effectively compensating for fine-grained parallax caused by viewpoint differences. Notably,

this progressive strategy is applied not only to the two modules but also within the second stage itself, allowing for hierarchical refinement from the global to the local. This design mimics the gradual nature of human-guided alignment while overcoming the limitations of traditional annotation methods.

Linear Transformation Module.

As shown in Fig. 2, the idea behind linear offset prediction is to amplify both channel-wise and spatial-wise differences between the two embeddings, making the misalignment more pronounced and easier to correct in subsequent steps. Firstly, we perform average pooling and maximum pooling along the channel dimension for $E^{Concat} \in \mathbb{R}^{2C \times L}$ from concatenated E^{vi} and E^{ir} , and re-concat them, having $E^{channel} \in \mathbb{R}^{4C}$ to feed into a channel-level MLP, followed by a sigmoid function to obtain the weight $W_c \in \mathbb{R}^{2 \times C}$:

$$W_{vi}^C, W_{ir}^C = Split(\sigma(MLP(E^{channel}))), \quad (1)$$

where W_{vi}^C and W_{ir}^C are the channel weighting vectors, $Split(\cdot)$ represents a feature separation operation, $\sigma(\cdot)$ is the Sigmoid activation function, $MLP(\cdot)$ stands for Multilayer Perceptron function.

In the meanwhile, another spatial-wise MLP consisting of convolution layers assembled with a Relu function, obtains the weight $W_s \in \mathbb{R}^{2 \times L}$ by E^{Concat} :

$$W_{vi}^S, W_{ir}^S = Split(\sigma(Conv(ReLU(Conv(E^{concat}))))). \quad (2)$$

For the enhanced thermal embedding, the procedure is as below, same as the thermal one:

$$E^{ir} = (W_{vi}^S * E^{vi} + W_{ir}^C * E^{vi})/2 + E^{ir}. \quad (3)$$

Furthermore, since the embeddings are derived from global modeling of image patches, and also incorporate multimodal learning, we leverage the difference between the two embeddings to predict three key values: the center coordinates x, y , height H , and width W of the thermal image. These values are predicted through a successive convolutional block and three prediction heads $\mathcal{H}$.

$$(x, y, H, W) = \mathcal{H}(Subtract(E^{ir}, E^{vi})), \quad (4)$$

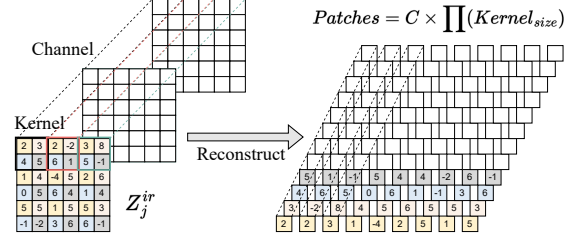


Fig. 3: Local Reconstruction uses the moved kernel to rearrange the spatial information to the channel dimension for calculation.

where $Subtract(\cdot)$ represents the element-wise subtract operation. These convolutional layers progressively refine the feature differences between the two modalities, allowing the network to capture meaningful transformation cues. Each branch in the prediction head uses a similar structure of adaptive average pooling, followed by a Conv1d layer that regresses the respective scaling values and the center coordinates.

Once the center coordinates and scaling factors are predicted, they are used to construct a transformation matrix ϕ that applies an affine transformation to the thermal image.

$$\phi = M_{grid(x, y, H, W)}, \quad (5)$$

where $M_{grid}(\cdot)$ represents constructing the matrix and mapping it to the grid. By applying these transformations, the network achieves weak alignment of the thermal and the RGB images, correcting for global misalignments in terms of scale and orientation as illustrated in Fig. 4(d). The process ensures that the thermal image is aligned with the RGB image at a coarse level, setting the stage for further refinement.

Warp Transformation Module.

To further address view offset issue, we propose a dense matching procedure $DM_{ir \rightarrow vi}$ to estimate the deformation field $\varphi_{ir \rightarrow vi}$ from linear transformed thermal images to RGB images. The dense match network consists of a series of convolutional layers designed for extracting modality-shared features F_1^{vi} , followed by the extraction of modality-independent features $F_j^{vi}, F_j^{ir}, j = 2, 3, 4$.

In the j -th match block, we firstly warp the source feature F_j^{ir} with the last estimated deformation field, and label it with Z_j^{ir} , the initial deformation field imported into the 3rd match is empty. To capture fine-grained pixel differences and accurately predict

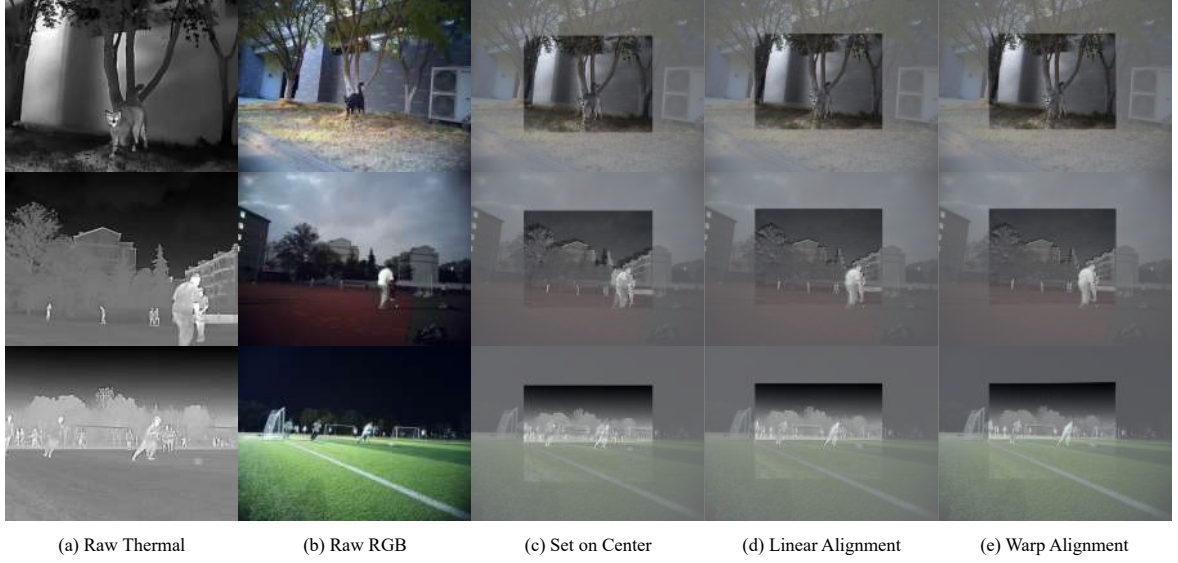


Fig. 4: Visualizations of some samples in each alignment stage on the UVT-VOD2024 dataset, where the first and the second columns show the raw images input into the whole model. The last three columns demonstrate that our aligner progressively corrects misalignment.

the deformation field, we locally reconstruct the Z_j^{ir} . Specifically, we use a kernel for sliding cropping, dividing fine-grained thermal features into overlapping patches, as shown in Fig. 3. This approach enables a more precise estimation of the local cross-modal matching. To manipulate the final shape corresponding to F_j^{vi} for subsequent operations, we set the appropriate kernel size, the padding size and the dilation size in local reconstruction. Then, we use a convolutional layer to extract local correlations between Z_j^{ir} and F_j^{vi} to find more detailed correspondences between different modalities:

$$Corr_j^{ir \rightarrow vi} = Conv(Subtract(F_j^{vi}, Z_j^{ir})). \quad (6)$$

The correlations are concatenated with the F_j^{vi} and the previous layer’s deformation field in the dimension. And another convolutional layer is used to model the current deformation field. Finally, the residual connection with the previous layer’s deformation field completes the transition matching between dense layers:

$$\varphi_j = Conv(Concat(Corr_j^{ir \rightarrow vi}, Z_j^{ir}, \varphi_{j-1}) + \varphi_{j-1}). \quad (7)$$

Each match block is used to sample features from the previous layer to achieve coarse alignment. This process is repeated through progressively shallower layers until the shallowest features are processed. Ultimately, a deformation field $\varphi_{ir \rightarrow vi}$ for image-level deformation is estimated. The entire process can be seen as a coarse-to-fine approach that progressively mines and resolves differences for alignment. The final thermal image can be achieved by re-sampling the unaligned thermal image with the estimated deformation field, as illustrated in Fig. 4(e). As shown, more edge details and the location of objects are further rectified.

3.4 ImageBind Guided Alignment Consistency

Given that the estimation process of dense match relies on features from two modalities rather than embeddings inferred from ImageBind, we constrain the alignment performance using the feature space of ImageBind. It is well understood that fully aligned embeddings from different modalities are closely aligned in ImageBind’s feature space. Meanwhile, the embeddings from ImageBind exhibit cross-modal emergent capabilities, where auxiliary modalities such as depth and IMU can be seamlessly integrated. These additional modalities provide implicit constraints that

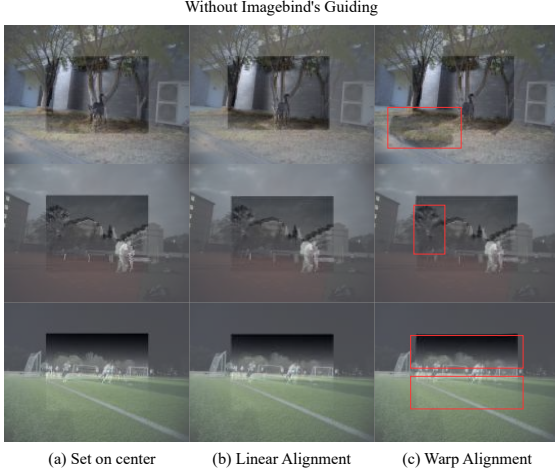


Fig. 5: Alignment samples from two-stage aligner in end-to-end detector without ImageBind’s guidance. The aligner instead introduces substantial distortions to large background regions as shown as the region in the red box from top to bottom. The deformations respectively focus on the ground, the tree and the sky.

further guide and enhance the learning of our two-stage aligner.

Therefore, after applying the warp transformation to the thermal image, we perform a center crop to avoid interference from the mask. The cropped image is then passed through the frozen encoder to obtain a new embedding, and the same is done for the RGB image. Using an alignment loss L_{Align} , we constrain the class tokens of both embeddings, a process that continually enforces the convergence of scene-level semantics from both modalities, pushing their distance in latent feature space as small as possible:

$$M_{sim} = O(E^{vi})^T \cdot O(E_{warp}^{ir}), \quad (8)$$

where M_{sim} is a matrix obtained from multiplication. Concretely, $O(\cdot)$ denotes selecting the first token (the class token) from the token sequence E^m (i.e., index 0). We apply L_{Align} on class tokens to enforce scene-level semantic convergence between modalities in the pretrained ImageBind latent space, while allowing patch tokens to remain flexible for geometric refinement in the progressive aligner:

$$L_{Align} = 1 - \text{Mean}(\text{Diag}(M_{sim})), \quad (9)$$

where $\text{Diag}(\cdot)$ represents the operation to select the elements of the main diagonal, $\text{Mean}(\cdot)$ represents the operation to take the mean value. In this way, each layer in dense match and the former cross-modal location are able to better learn the critical steps of alignment. Under the alignment consistency guided by ImageBind, the aligned thermal image is progressively encouraged to produce embeddings that closely approximate those of the corresponding RGB image.

When relying solely on the supervision imposed by the detection task, the aligner struggles to converge, making it difficult to capture semantically meaningful correspondences. This issue is illustrated in Fig. 5, which shows the alignment performance of the two-stage aligner in the absence of guidance from ImageBind. During the Linear Transformation stage, the alignment remains inconspicuous, appearing to degrade into simple data augmentation like Mixup. In the subsequent Warp Transformation stage which is originally intended for fine-grained refinement, the aligner instead introduces substantial distortions to large background regions. This phenomenon highlights the critical role of ImageBind guidance, which not only depends on direct alignment between thermal and RGB modalities but also benefits from emergent cross-modal pairings involving other modalities such as depth and IMU.

3.5 Thermal-to-RGB Fusion

Owing to the inherently limited receptive field of thermal imaging, directly fusing with RGB modalities would compromise global contextual information in the RGB data, even following spatial alignment. Crucially, the principal advantages of thermal imaging reside in its high luminance contrast and well-defined boundary characteristics, which are particularly valuable for object delineation under low-visibility conditions.

Building upon this framework, we implement a homologous attention mechanism where thermal features serve as both queries and keys, while RGB features operate as values. This design enables the attention map to preferentially emphasize foreground objects in the thermal modality. Additionally, homologous attention often introduces most of features from the value, which goes along with our intent to leverage the positional priors from the thermal image while preserving the detailed features of the RGB image. The inputs to the T2RF module consist of the RGB modality feature map and the thermal feature map, we

compute a cross-modal similarity via an element-wise inner product:

$$Sim = \frac{f_{ir} \cdot f_{vi}}{\|f_{ir}\|_2 \|f_{vi}\|_2}, \quad (10)$$

where $f_{ir} \in \mathbb{R}^{C \times i}$, $f_{vi} \in \mathbb{R}^{C \times i}$, and $i \in \{1, \dots, hw\}$, and apply it to bridge the cross-modal local feature to promote the later attention:

$$\begin{aligned} f'_{ir} &= ReLU(f_{ir} + Sim \odot f_{vi}), \\ f'_{vi} &= ReLU(f_{vi} + Sim \odot f_{ir}). \end{aligned} \quad (11)$$

Then we learn self-attention affinity for thermal feature via:

$$affinity_{ij} = Softmax(g(f'_{ir})^i \cdot g(f'_{ir})^j), \quad (12)$$

where $g(\cdot)$ denotes flatten on hw and $i, j \in \{1, \dots, hw\}$ are the indices. The self-attention affinity identifies image patches with similar characteristics and corresponds to saliency representations for foreground in the thermal feature. We then apply a matrix multiplication with the attention maps to enhance the RGB feature with more location-aware information:

$$f_{vi}^* = ReLU(Reshape(\alpha(f_{vi} \otimes Affinity)) + f_{vi}), \quad (13)$$

where we introduce a factor α to adaptively learn the fusion effect from thermal modality to the RGB one. The original RGB features are preserved through a residual connection to ensure their integrity.

3.6 Temporal Encoder

To balance generality across both video and image datasets, and to control the overall task complexity, we adopt a deliberately light-weight temporal design. Since IPTNet already tackles the challenging alignment-free multi-modal correspondence under raw unaligned inputs, we keep temporal modeling efficient by directly feeding multiple frames into a Deformable-DETR-style encoder to aggregate spatio-temporal context, without introducing an additional heavy temporal branch. Given a clip of T frames $\{I_t\}_{t=1}^T$, our model produces multi-scale fused feature maps $\{x_t^l \in \mathbb{R}^{C \times H_l \times W_l}\}_{l=1}^L$ for each frame t . We regard the collection $\mathcal{X} = \{x_t^l\}_{t=1..T, l=1..L}$ as the memory to be encoded. For each query element q with content feature z_q and normalized reference

point $\hat{p}_q \in [0, 1]^2$, we extend multi-scale deformable self-attention to the temporal dimension:

$$\begin{aligned} \text{T-MSDeformAttn}(z_q, \hat{p}_q, \mathcal{X}) = \\ \sum_{m=1}^M \mathbf{W}_m \sum_{t=1}^T \sum_{l=1}^L \sum_{k=1}^K A_{tlqk}^m \cdot \\ \mathbf{W}'_m \mathbf{x}_t^l(\phi_l(\hat{p}_q) + \Delta \mathbf{p}_{tlqk}^m), \end{aligned} \quad (14)$$

where m indexes the attention head, k indexes the sampled key, and $\phi_l(\cdot)$ maps normalized coordinates to level- l feature coordinates. $\Delta \mathbf{p}_{tlqk}^m \in \mathbb{R}^2$ and $A_{tlqk}^m \in [0, 1]$ denote the sampling offset and attention weight, predicted from z_q and normalized by $\sum_{t,l,k} A_{tlqk}^m = 1$. Sampling at fractional locations uses bilinear interpolation. Eq. (14) preserves the efficiency of deformable attention by attending to a small set of locations per query, and keeps the temporal module lightweight. In practice, we use a small $T = 2$ for efficiency and fair adaptation across different datasets: we enable the temporal encoder for video datasets, while we disable it for image datasets. The encoded memory is then passed to the detector decoder to output the detection result for the current frame.

3.7 Overall Loss Function

Eventually, we sent f_{vi}^* and its reference frame into DINO to finish detection. The training loss of our model follows the DETR-like detectors, which is defined as:

$$L = L_{cls} + L_{box} + L_{dn} + L_{Align} + L_{MSE},$$

where L_{cls} is the IoU-aware classification loss following DINO, L_{box} is composed of L1 loss and generalized IoU loss for bounding box regression, L_{dn} is the loss for denoising training following DINO, L_{Align} is our Alignment Consistency and L_{MSE} is a common measure of feature similarity function acting directly on two modalities' multiscale features. In addition, we also calculate the L_{cls} , L_{box} and L_{dn} of each decoder layer as the auxiliary optimization loss. L_{Align} and L_{MSE} are applied at intermediate stages of the model to learn alignment, while the other losses are applied to the decoder head. All losses have the same configured weights except for their positions.

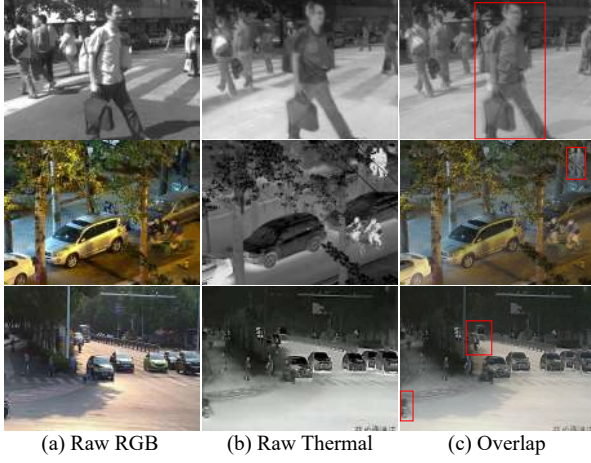


Fig. 6: The visualization examples of the misalignment. From top to bottom, these examples come from CVC14, LLVIP and VT-VOD50, where (c) displays misaligned pixels' shift in red box.

4 Experiments

4.1 Datasets and Evaluation Metric

UVT-VOD2024.

The UVT-VOD2024 Benchmark [49] contains 174 videos of varying sizes, totaling 30,494 pairs of RGBT images, where 118 videos are selected for training and 56 videos are for evaluation. The resolution of RGB image is 1600×1200 and the resolution of thermal image is 640×480 in the dataset, bringing the challenge that need us to carry on the cross-modality location and mitigate the general view difference from RGBT sensors. Another important issue is that only the annotations of RGB video object detection results exist, which requires us to make the most of the thermal information under misalignment scenarios.

VT-VOD50.

The VT-VOD50 Benchmark [48] consists of 50 pairs of RGBT videos and a total of 9449 RGBT image pairs, where 38 videos are for training and 12 videos are used for evaluating. The videos in the dataset have various resolutions, including 640×368 , 680×404 , 720×576 , 720×404 , 1920×1080 , and other aspect ratios, all of which have been manually aligned. As mentioned before, even manual alignment might have weak misalignment, so objects at the edge of the scene are not aligned well. We take experiments on this basis to verify the robustness of our method. Additionally,

we explore the performance gap between our fusion approach and other methods on this benchmark.

CVC-14.

The CVC-14 dataset [22] has 7,085 RGB (grayscale) and thermal image pairs for training, where 3,695 pairs are captured during daytime and 3,390 pairs during nighttime. The test set is composed of 1,433 frames. Since the cameras are not well calibrated, the annotations in CVC-14 are individually provided in each modality. There is no resolution difference like in UVT-VOD2024. However, the CVC-14 dataset suffers from the severe position shift between RGBT image pairs.

LLVIP.

LLVIP [27] is a dataset with paired RGB-thermal data, captured in the low-light condition. It has 15,488 pairs of aligned images, with 42,437 annotated pedestrians total. Most of them are taken under dark scenarios by surveillance camera. The quality of original RGB and thermal images in the LLVIP dataset is significantly higher than that of the CVC-14 dataset. The RGB and thermal images have the resolution of 1920×1080 and 1280×720 , respectively. LLVIP avoids the edge misalignment by cropping, but this misalignment situation still remains in the final registration data set, which is important for testing the methods based on aligning multimodal data.

Evaluation Metric.

For UVT-VOD2024, VT-VOD50 and LLVIP, we employ the commonly used average precision metrics (AP_{iou}) under different IoU thresholds as evaluation metrics. Specifically, we select $AP_{0.5}$ known as AP_{50} and AP as metrics for comparison. The AP metric represents the mean AP_{iou} , whose IoU ranges from 0.50 to 0.95 with a stride of 0.05. For CVC-14, we employ the log miss rate (MR^{-2}) averaged over the false positive per image (FPPI) with a range of $[10^{-2}, 10^0]$ for evaluation. A lower MR^{-2} indicates better performance. Despite manual alignment, some misalignments still exist in these benchmarks due to real-world challenges, as illustrated in Fig. 6.

4.2 Implementation Details

The IPTNet is developed using DINO architecture as the detector [58]. We load the pretraining weight of DINO on the COCO dataset for 12 epochs. We employ ResNet50 as the backbone for both the thermal and the

RGB branches to extract multi-level features as similar as DINO. Both Transformer encoder and decoder have six layers. In our decoder, the number of attention heads, sampling points, and object queries are set to be $H = 8$, $K = 4$ and $N = 900$. The experiments are carried out on a single NVIDIA A100 GPU with 40 GB of memory for 50 epochs. We implement our algorithm with the PyTorch framework with Python and the Adamw optimizer with weight decay of 0.0001. The initial learning rate is set to 0.0001. The input size of images in our model is only factor scaled in the preprocessing stage while the original ratio remains unchanged on the specific benchmark. We train our model with the entire video sequence rather than randomly sampled part of the sequence. For each frame, it is always modeled with the next closest key frame in the sequence. IPTNet uses approximately 23 GB GPU memory during training and approximately 18 GB during inference.

The official ImageBind inference API typically outputs a single global embedding by directly indexing the class token for each modality. In our implementation, we keep all ImageBind encoders frozen and do not change the architecture. We only retrieve the encoder output (the full token sequence) before the final class-token indexing:

$$E^m = [e_{cls}^m; e_1^m; \dots; e_{L-1}^m] \in \mathbb{R}^{L \times C}, \quad (15)$$

where e_{cls}^m denotes the class token and the remaining tokens correspond to patch embeddings.

ImageBind’s thermal encoder expects a single-channel thermal input. For datasets where pseudo-thermal images are stored as three channels, we convert them to one channel by weighted averaging:

$$I^{ir} = \sum_{c=1}^3 w_c I_c^{ir}, \quad w_c = \frac{1}{3}. \quad (16)$$

We do not modify ImageBind’s input layer. After conversion, we keep the default ImageBind normalization statistics for the thermal modality unchanged (mean = 0.5, std = 0.5), so that the pretrained priors are preserved to the maximum extent.

4.3 Comparison with State-of-the-Art Methods

Evaluation on UVT-VOD2024 & VT-VOD50.

We compare our model on two RGBT VOD benchmark datasets UVT-VOD2024 and VT-VOD50

with several state-of-the-art OD/VOD methods. Both datasets are specifically designed for video tasks, where the former explicitly presents the significant misalignment challenge, while the latter is manually aligned, serving as a robust test against slight misalignments.

As there are currently few end-to-end methods specifically addressing misaligned RGBT object detection, we compare our approach with state-of-the-art video object detection and image object detection methods. To integrate multimodal information, we adopt pixel-wise addition as a baseline fusion strategy, where most results are from [49].

As demonstrated in Table 1, IPTNet not only resolves the critical misalignment issues in UVT-VOD2024, but also shows a more effective fusion effect, achieving state-of-the-art performance on both datasets. Under UVT-VOD2024, our IPTNet outperforms MDLNet by 5.9. When averaging AP and AP50 for an overall comparison, IPTNet achieves an approximate 13%. MDLNet [49] addresses the misalignment issue through a series of prior-based patch-level interactions. While this approach mitigates misalignment to some extent, it remains a relatively coarse alignment method. In contrast, our alignment strategy strives for precise pixel-level alignment, enabling our approach to significantly outperform this benchmark. This performance comparison illustrates that our method can well align and fuse the two modalities. Notably, the pre-aligned VT-VOD50 dataset contains slight and imperceptible misalignments. Our method’s superior performance over the SOTA method MDLNet demonstrates its capability to correct even slight misalignments, which are often overlooked by other models. Regarding efficiency, IPTNet achieves 10.2 FPS with 101.1M optimized trainable parameters and 512 GFLOPs. While our method is not designed to be the fastest, these results indicate a reasonable trade-off between accuracy and efficiency for alignment-free RGBT VOD. The computation is comparable to representative video-based baselines while delivering significantly better detection accuracy.

Evaluation on CVC-14.

We further compare our method with numerous state-of-the-art approaches on the CVC-14 dataset, that is also the RGBT dataset with misaligned data. Unlike these methods which typically rely on the pretraining with large-scale KAIST[13] dataset, we directly fine-tune the COCO-pretrained weights on CVC-14.

Table 1: Comparison with the state-of-the-art OV/VOD methods on the UVT-VOD2024 and VT-VOD50 dataset. We highlight the best results in **bold**. The “—” indicates that the measurement conditions are not met or that the result cannot be obtained.

Methods	Backbone	Type	UVT-VOD2024		VT-VOD50		FPS	Params(M)	FLOPs(G)
			AP ₅₀ (%)	AP(%)	AP ₅₀ (%)	AP(%)			
<i>Image-based detectors</i>									
YOLOV3 [38]	Darknet53	Image	25.6	13.5	33.9	17.4	69.9	103.7	283
YOLOV5_M [28]	CSPDarknet53	Image	23.9	12.5	—	—	294.1	25.1	64.4
CFT [37]	CFB	Image	6.7	2.4	42.5	18.9	222.2	73.7	—
YOLOX_L [19]	Darknet53	Image	16.3	—	—	—	104.8	54.2	155.8
YOLOV6_M [30]	EfficientRep	Image	22.7	12.1	—	—	169.5	52.0	161.6
YOLOV7 [45]	CSPDarknet53	Image	23.0	10.4	37.7	16.5	294.1	36.5	103.3
YOLOV9-C [46]	GELAN	Image	27.3	14.5	49.1	26.9	99.0	25.5	103.7
YOLOV10-M [44]	CSPNet	Image	17.1	8.7	46.2	25.2	210.0	16.5	64.0
EfficientDet [41]	EfficientNet	Image	20.2	8.8	—	—	87.0	20.0	100
TOOD [17]	ResNet-50	Image	15.9	7.3	36.3	19.0	25.8	32.0	199
Deformable DETR [69]	ResNet-50	Image	7.7	2.9	42.5	23.3	20.7	41.1	197
RT-DETR [62]	ResNet-50	Image	17.0	7.9	40.2	21.6	—	42.7	130.5
DINO [58]	ResNet-50	Image	29.4	13.7	47.4	25.9	16.7	47.7	274
AlignDETR [2]	ResNet-50	Image	21.1	9.2	—	—	12.9	47.5	235
DDQ DETR [60]	ResNet-50	Image	21.1	9.1	48.3	26.5	13.0	48.3	275
DiffusionDet [9]	ResNet-50	Image	21.4	9.6	46.9	25.1	—	—	—
<i>Video-based detectors</i>									
DFF [72]	ResNet-50	Video	9.2	3.9	33.5	14.1	40.4	62.1	24.9
FGFA [71]	ResNet-50	Video	16.7	—	35.1	15.8	9.0	64.5	41
RDN [15]	ResNet-50	Video	16.9	—	40.0	—	11.3	—	—
SELSA [50]	ResNet-50	Video	12.6	4.6	39.4	17.4	10.5	—	—
MEGA [11]	ResNet-50	Video	15.4	—	27.8	—	16.2	—	—
Temporal ROI Align [21]	ResNet-50	Video	11.1	3.9	38.0	17.0	5.1	—	—
CVA-Net [32]	ResNet-50	Video	16.4	6.4	39.7	19.7	6.9	41.6	548.1
STNet [36]	ResNet-50	Video	15.7	6.5	38.4	18.4	5.0	41.6	752.3
EINet [48]	Darknet53	Video	20.7	—	46.3	24.0	204.2	11.6	78.2
MDLNet-X [49]	CSPDarknet53	Video	35.2	18.4	—	—	19.5	189.4	1038.2
<i>Ours</i>									
IPTNet(Ours)	ResNet-50	Video	41.1	19.9	60.1	35.0	10.2	101.1	512

Although CVC-14 provides ground-truth annotations for both RGB and thermal data, we only utilize the annotations of RGB modality, treating the thermal modality purely as auxiliary information. This configuration better reflects the superiority of our method in handling misalignment, that is our method is more suitable for real-world scenarios compared to other approaches. It is worth noting that we do not use a temporal encoder to be consistent with other methods. From the Table 2, we can observe that IPTNet performs well under “All”, “Day” and “Night” settings. Despite the smaller scale and misaligned situation of the CVC-14 dataset, our approach achieves notable detection performance. Note that our model can obtain a lower miss rate compared with other models, especially in the “All” scenario, due to the

misalignment situation in CVC-14 is not as complicated as UVT-VOD2024. Except for method [8], which is pre-trained on the KAIST dataset and thus shows a particular advantage under both day and night conditions, our method still performs comparably with only a small gap.

Evaluation on LLVIP.

For LLVIP dataset, the results are provided in Table 3. Thanks to the wide availability of this dataset, we have annotated the source of each comparison method. Since LLVIP is a dataset of night scenes, a good fusion strategy like T2RF that introduces less irrelevant features determines the upper limit of the final detection result. The LLVIP dataset presents a challenging scenario, for which the performance of UniRGB-IR [54] in the AP50 metric is already nearly perfect. Despite

Table 2: Comparison with state-of-the-art methods on CVC-14. Lower is better for MR^{-2} at IoU=0.5. “*” indicates fine-tuning based on KAIST pretraining.

Methods	$MR^{-2} \downarrow$ (IoU=0.5)		
	All	Day	Night
MACF [35]	60.1	61.3	48.2
Choi et al. [12]	47.3	49.3	43.8
Halfway Fusion [35]	37.0	38.1	34.4
Park et al. [35]	31.4	31.8	30.8
AR-CNN [59]	22.1	24.7	18.1
MCHE-CF [31]	21.3	24.0	17.5
MLPD [29]	21.3	24.2	18.0
MBNet* [65]	21.1	24.7	13.5
AAANet-Faster R-CNN* [8]	<u>17.9</u>	22.6	10.9
MS-DETR [53]	20.0	23.0	15.1
CAMF [57]	20.2	22.8	12.4
IPTNet (Ours)	16.9	23.8	<u>11.4</u>

Table 3: Comparison with state-of-the-art methods on LLVIP.

Methods	Publication	AP $\uparrow$	AP ₅₀ $\uparrow$
PoolFuser [5]	AAAI 2023	38.4	80.3
DetFusion [40]	ACM MM 2022	—	80.7
ProbEn [10]	ECCV 2022	51.5	93.4
DIVFusion [43]	Information Fusion 2023	52.0	89.8
LENFusion [7]	TIM 2024	53.0	81.6
DM-Fusion [55]	TNNLS 2023	53.1	88.1
CAMF [42]	TMM 2024	55.6	89.0
MoE-Fusion [3]	ICCV 2023	—	91.0
MetaFusion [61]	CVPR 2023	56.9	91.0
TFNet [14]	TITS 2023	57.6	—
DDFM [63]	ICCV 2023	58.0	91.5
DCMNet [51]	ACM MM 2022	58.4	—
CSAA [4]	CVPR 2023	59.2	94.3
Diff-IF [56]	Information Fusion 2024	59.5	93.3
YOLO-Adaptor [18]	TIV 2024	—	<u>96.5</u>
CALNet [26]	ACM MM 2023	63.4	—
UniRGB-IR [54]	ACM MM 2025	63.2	96.1
IPTNet (Ours)	—	65.9	96.7

this, our method establishes a new state-of-the-art. It achieves a modest 0.6% improvement in AP50, but more significantly, a substantial 2.7% gain in the comprehensive AP metric. This result demonstrates our method’s capacity to mitigate the slight non-linear misalignments, that persist even after manual alignment. By resolving these discrepancies, our model effectively extracts fine-grained, complementary information from the fused modality.

4.4 Ablation and Analysis

To investigate the effectiveness of the progressive alignment and the T2RF, we conduct some ablation studies. For simplicity, we only conduct experiments on the UVT-VOD2024 dataset and show the results in AP and AP50.

4.4.1 Effectiveness of The Alignment Components.

We systematically remove or modify each key component and observe the impact on the model’s performance based on T2RF module to verify the effectiveness of the proposed progressive alignment. The T2RF mechanism is designed with the goal of leveraging the advantages of the thermal modality after alignment, particularly in scenarios where the thermal modality has a low receptive field. Therefore, modifying either of two-stage aligner before the T2RF will significantly affect the final result. As observed in Table 4, the fusion performance without any explicit alignment is modest. However, when we simply place the thermal modality in the center of the RGB image as a default setup, T2RF yields a significant accuracy improvement, which is what the resolution gap causes. Next, we explore replacing the center placement with the Linear and the Warp Transformations. Linear Transformation achieves effective global positioning and slight adjustment in scale, leading to a 0.5% improvement compared to the second row. On the other hand, using Warp Transformation alone without preliminary alignment except center operation, struggles to achieve finer-grained adjustments, resulting in the performance comparable to Linear Transformation. The removal of Alignment Consistency supervision leads to a noticeable decline in alignment performance, indicating its critical role in guiding effective cross-modal correspondence. Finally, combining both the transformations in a progressive alignment strategy delivers a remarkable improvement, demonstrating the effectiveness of this gradual refinement approach.

4.4.2 Exploring Stronger Emergent-based Alignment Consistency.

In the main setting, our alignment consistency provides a concise and stable emergent supervision in the frozen ImageBind latent space, which is shown to be critical for effective correspondence learning. We further explore two stronger variants that aim to exploit more potential of ImageBind-guided consistency.

Table 4: Ablation of progressive alignment components on UVT-VOD2024 (with T2RF). “Center” denotes directly placing the thermal image at the center of the RGB image before transformation. Δ is computed w.r.t. Setting S1.

Setting	Progressive Alignment Components				Detection Performance		Gain vs. S1	
	Linear Transform	Warp Transform	Alignment Consistency	T2RF	AP $\uparrow$	AP $_{50} \uparrow$	Δ AP	Δ AP $_{50}$
S1	None	$\times$	$\times$	$\checkmark$	18.1	36.6	+0.0	+0.0
S2	Center	$\times$	$\times$	$\checkmark$	18.5	37.7	+0.4	+1.1
S3	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	18.6	37.9	+0.5	+1.3
S4	Center	$\checkmark$	$\checkmark$	$\checkmark$	17.7	37.9	-0.4	+1.3
S5	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	18.3	37.6	+0.2	+1.0
S6	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	19.9	41.1	+1.8	+4.5

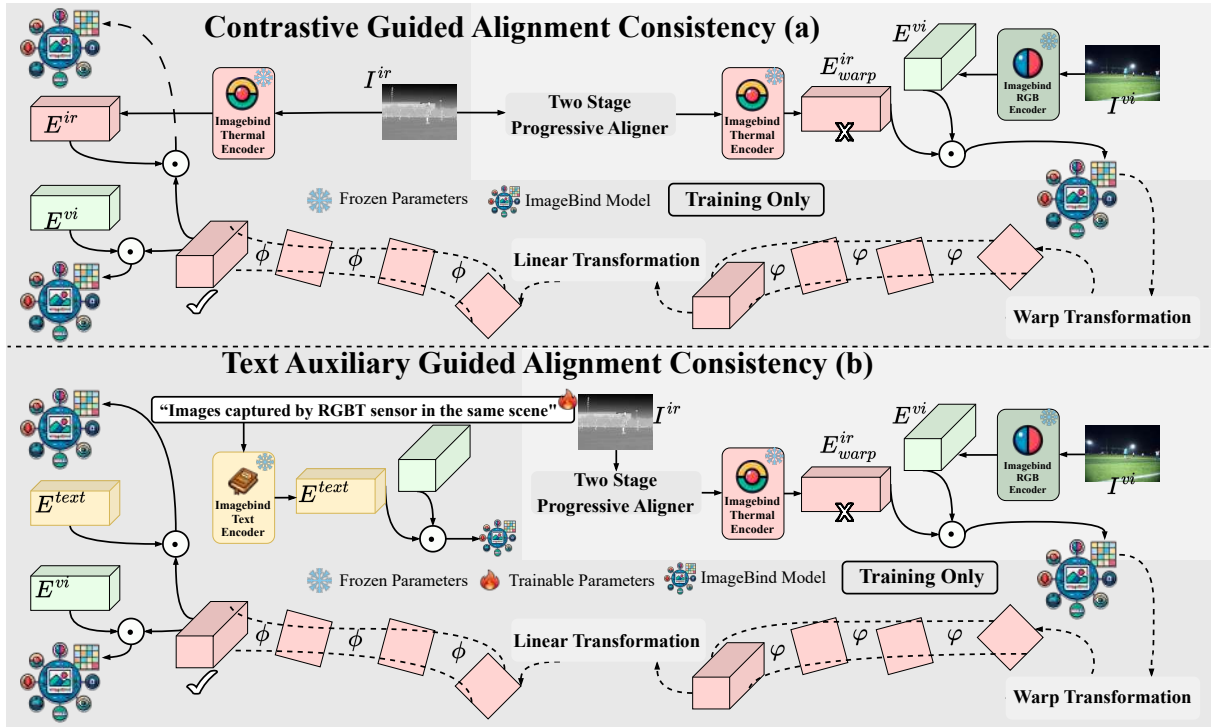


Fig. 7: (a) Contrastive Guided Alignment Consistency shows how the raw and warped thermal embedding are deployed in dataflow. (b) Text Auxiliary Guided Alignment Consistency shows how the trainable text embedding works in dataflow.

Contrastive Guided Alignment Consistency.

Besides pulling the warped thermal embedding closer to the RGB embedding, we additionally encourage it to be closer than the raw (unaligned) thermal embedding for each frame, as illustrated in Fig. 7(a). Concretely, we impose a simple ranking constraint

between the similarities:

$$\mathcal{L}_{con} = \max(0, \text{sim}(o^{vi}, o^{ir}_{raw}) - \text{sim}(o^{vi}, o^{ir}_{warp})), \quad (17)$$

where o^{vi} , o^{ir}_{warp} , and o^{ir}_{raw} are the ImageBind class tokens, $\text{sim}(\cdot, \cdot)$ denotes cosine similarity.

Table 5: Performance comparison of different data sources for linear transformation matrix prediction on UVT-VOD2024.

Source of linear matrix prediction	AP $\uparrow$	AP $_{50}\uparrow$
Backbone feature	15.7	37.5
Backbone feature*	14.3	33.6
ImageBind embedding	19.0	39.5
LanguageBind embedding	19.2	40.5
Enhanced ImageBind embedding	19.9	41.1

Text Auxiliary Guided Alignment Consistency.

We also introduce a text modality as an auxiliary anchor by initializing a prompt: “Images captured by RGBT sensor in the same scene”. We encode it once using the frozen ImageBind text encoder to obtain a text embedding o_0^{txt} . Then, instead of updating the text encoder, we treat the text embedding itself as a learnable parameter and allow it to be dynamically updated during training, while keeping the ImageBind text encoder fully frozen. The detailed process details are illustrated in Fig. 7(b). Given a prompt embedding o^{txt} , we encourage both RGB and warped thermal to be consistent with this anchor:

$$\mathcal{L}_{txt} = (1 - \text{sim}(o^{vi}, o^{txt})) + (1 - \text{sim}(o_{warp}^{ir}, o^{txt})). \quad (18)$$

We report the results in Table 6. Interestingly, although both variants are conceptually reasonable, they do not outperform the original concise consistency, because the raw thermal frame still shares strong scene semantics with RGB, enforcing an explicit repulsion at the global ImageBind token level may over-penalize semantically correct pairs, and weakening useful cross-modal correspondence cues for detection. Similarly, the text anchor provides a global semantic prior, but it is not geometry-aware and does not reflect frame-specific layout variations. Thus it may bias the embeddings toward generic semantics and reduce the flexibility needed by the fine-grained warping stage. When combining both constraints, the regularization becomes stronger and the degradation is more evident, indicating that overly strong semantic constraints may conflict with localization-sensitive learning. Overall, the concise alignment consistency in Eq. 8,9 offers a better trade-off between stable emergent supervision and geometric adaptability.

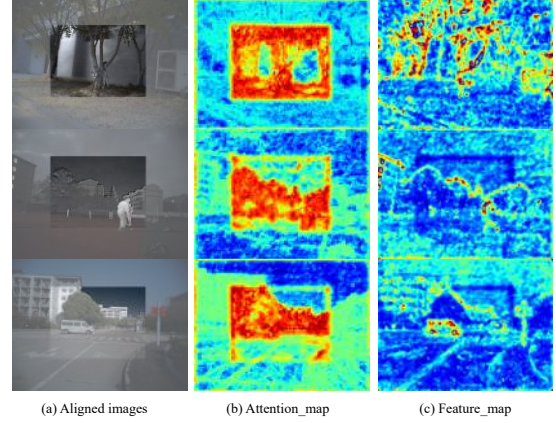


Fig. 8: Visualization of attention map and feature map of mid layer in T2RF module, where (a) shows the aligned images, (b) shows the region the T2RF attention focuses on, (c) shows the fused feature map for detection. The visualization of the intermediate results demonstrates that the model takes full advantage of the aligned thermal image. One can zoom in for more details.

4.4.3 Effectiveness of Different Sources for Linear Transformation.

To further evaluate the outstanding performance of ImageBind in our progressive alignment framework, we analyze the data sources when predicting the linear transformation matrix in the initial step. The type of data sources is defined as follows:

- **BackBone Feature:** In this setting, we use ResNet-50 as the backbone for both modalities instead of the encoder from ImageBind. The difference is calculated based on the features extracted by the backbone, which are then used for prediction.
- **BackBone Feature*:** This is a stricter variant without ImageBind guidance. Compared with the full IPTNet, it first replaces ImageBind embeddings with ResNet-50 backbone features for linear transformation prediction, as in the “BackBone Feature” setting. It further removes the ImageBind-guided Alignment Consistency loss. All other components are kept unchanged. Therefore, the aligner receives neither ImageBind embeddings nor ImageBind-based supervision, and is optimized only by the detection losses.

Table 6: Exploring stronger emergent supervision for alignment consistency on UVT-VOD2024. All settings use the same progressive aligner and T2RF as the full model.

Method Variant	Consistency Supervision			Performance	
	Base Consistency	Contrastive (Raw IR)	Text Anchor	AP $\uparrow$	AP $_{50}$ $\uparrow$
Alignment Consistency $\mathcal{L}_{Align}$	✓	✗	✗	19.9	41.1
+ Contrastive $\mathcal{L}_{con}$	✓	✓	✗	19.5	40.4
+ Text Anchor $\mathcal{L}_{txt}$	✓	✗	✓	19.4	40.2
+ Both	✓	✓	✓	19.1	39.7

- **ImageBind Embedding:** Here, the difference is computed directly from the embeddings generated by the ImageBind encoder, simple and pure.
- **Enhanced ImageBind Embedding:** This is the final configuration determined in this study, where embeddings are further enhanced for alignment.
- **LanguageBind Embedding:** In this configuration, we replace ImageBind with a language-centric alignment model called LanguageBind [67], aligning embeddings around the language modality.

These variations allow us to systematically assess the impact of different embedding strategies on the progressive alignment process. It can be observed from Table 5 that the model trained on the setting of Backbone feature can bring the tremendous drop of AP without introducing any additional module, which means that it can not be satisfied to calculate the global position even if we extract modality-specific features. This is because a normal backbone does not contain any prior of cross-modal alignment. The “BackBone Feature” setting removes ImageBind embeddings from the linear transformation prediction but still preserves the ImageBind-guided Alignment Consistency loss. Therefore, it only evaluates whether detector backbone features can replace ImageBind embeddings as the source of global alignment priors. In contrast, “BackBone Feature*” further removes the ImageBind-guided Alignment Consistency loss, making it a stricter setting without ImageBind guidance. Its performance drop indicates that, when no prior-rich shared embedding space is available, the aligner has difficulty in learning meaningful cross-modal correspondence from detection losses alone. This result further demonstrates the importance of ImageBind guidance in the alignment-free RGBT setting.

Once we simply use ImageBind Embedding setting, we can easily see the improvement of AP since the alignment prior is introduced. Not only does ImageBind use strictly aligned data for training, but the class token representing the whole image information makes it easier to predict the global relationship. The decision to enhance the embeddings stems aims to fully utilize the patch embeddings rather than relying solely on the class tokens. Since the encoder operates directly on raw images, the patch embeddings are not inherently aligned, and the channel dimensions between RGB and thermal images differ. By applying straightforward spatial and channel-wise enhancements, we can better capture the global difference through patch embeddings. As shown in the Table 5, introducing the enhanced setting leads to a notable improvement compared to configurations without this enhancement, demonstrating its effectiveness in refining the alignment process.

Table 5 separates the contribution of the foundation embedding prior from our enhancement. Replacing backbone features with raw ImageBind embeddings yields a large improvement, indicating that ImageBind indeed introduces a useful cross-modal anchoring prior, while our enhancement further improves performance by adapting patch-level statistics and mitigating modality-specific discrepancies to better exploit this prior.

We explicitly choose ImageBind due to its prior on aligned data. The Bind series of models has consistently advanced in classification and cross-modal retrieval tasks, with LanguageBind achieving stronger semantic understanding by shifting the alignment center to the language modality. To investigate whether semantics or spatial alignment plays a more critical role in cross-modal alignment, we replace the core Bind model and conduct the experiment. This experiment reaffirms a core premise of our work. First, the alignment priors learned from spatially-paired multi-modal data are indispensable for achieving effective



Fig. 9: Similar to other classic detectors, we visualized the detection results of IPTNet in several scenarios in VT-VOD50.

Table 7: Effect of different local reconstruction strategies for deformation prediction on UVT-VOD2024.

Type of local reconstruction	AP $\uparrow$	AP $_{50}$ $\uparrow$
None	19.6	39.8
Conv	19.2	39.7
Average Pooling	17.4	35.9
Max Pooling	16.0	33.6
Local Reconstruction	19.9	41.1

alignment. Second, it is precisely this inherent pairing that enables our model to perform end-to-end alignment, completely eliminating the need for additional annotations.

We note that raw LanguageBind can be slightly stronger than raw ImageBind, which is expected. LanguageBind can be viewed as a second-generation Bind model that builds upon ImageBind initialization and further improves the architecture and expands the training data, often resulting in stronger semantic alignment. We include LanguageBind for completeness, but we adopt ImageBind as our base because it is image-centric and natively covers the thermal modality, which better matches our objective of aligning thermal observations into the RGB coordinate frame. Importantly, the enhanced ImageBind configuration achieves the best performance, indicating that ImageBind prior and our adaptation are complementary.

4.4.4 Influence of Local Reconstruction in Deformation Prediction.

We design Local Reconstruction to uncover more pixel-level discrepancies, thereby achieving precise alignment. From a receptive field perspective, Local

Reconstruction is similar to convolution and pooling operations, but its functionality is more similar to "convolving without aggregation." Rather than discarding spatial information, it shifts this information to the channel dimension for further refinement. Table 7 provides an ablation study on the different substitutes of Local Reconstruction. To validate its effectiveness, we compared Local Reconstruction against similar operations, including convolution, max pooling, average pooling, and without any kernel operations. This comparison highlights the unique advantages of Local Reconstruction in capturing and addressing pixel-level misalignment.

The first row configuration directly uses the feature maps for progressive prediction of the deformation field. While this achieves reasonable results, the lack of diversity in channel information limits its capacity to capture the subtle details needed for precise pixel-level alignment. Using a convolution with the same kernel size as Local Reconstruction introduces some degree of local feature extraction. However, convolution inherently aggregates local features, which is counterproductive for pixel-level alignment. This aggregation causes a loss of fine-grained spatial information, explaining the marginal improvement over the baseline but still falling short compared to Local Reconstruction. Both pooling methods result in a significant degradation of performance, particularly in AP, due to the inherent nature of pooling operations, which discard spatial information. Average pooling smooths features indiscriminately, while max pooling retains only the strongest responses, both of which are detrimental to preserving the nuanced details required for precise alignment. The superior performance of Local Reconstruction can be attributed to its ability to "convolve without aggregation." By transforming spatial information into the channel dimension,

Table 8: Layer-wise study of T2RF insertion on UVT-VOD2024. “Layers” denotes the number of backbone stages equipped with T2RF. Δ_{S1} is the absolute gain over S1 (L1 only), and Δ_{prev} is the incremental gain over the previous setting.

Setting	T2RF Layers	Layers	Detection Performance		Gain (AP / AP ₅₀)	
			AP $\uparrow$	AP ₅₀ $\uparrow$	Δ_{S1}	Δ_{prev}
S1	L1	1	18.8	39.9	+0.0 / +0.0	–
S2	L1, L2	2	19.6	40.8	+0.8 / +0.9	+0.8 / +0.9
S3	L1, L2, L3	3	19.5	40.5	+0.7 / +0.6	-0.1 / -0.3
S4	L1, L2, L3, L4	4	19.9	41.1	+1.1 / +1.2	+0.4 / +0.6

it preserves pixel-level details while simultaneously enhancing the diversity of the feature representation. This design avoids the pitfalls of feature aggregation seen in convolution and pooling, allowing for more effective refinement of the deformation field.

4.4.5 Influence of T2RF on Multi-Scale.

We investigate the impact of T2RF when applied at different feature levels. Specifically, we extract four hierarchical feature maps from the aligned images, with levels 1 through 4 representing shallow to deep layers, in order to evaluate the model’s perception of objects at varying scales. At shallow layers, the disparity in receptive fields between the two modalities poses the major issue that T2RF is designed to address. Therefore, we begin the fusion at level 1 and progressively incorporate deeper features. At deeper layers, the effect of receptive field gap becomes negligible. Nevertheless, our results as in Table 8 show that the homology-driven mechanism of T2RF exhibits strong generalization ability, consistently improving the performance even when deeper features with smaller spatial resolutions are introduced.

4.5 Generalization under Synthetic Misalignment

Although real-world RGBT benchmarks (e.g., UVT-VOD2024) already contain non-negligible unalignment, we further evaluate the generalization of alignment robustness by constructing synthetic disturbances based on the original test set. Specifically, for each test video, we keep the RGB frames unchanged and apply an additional geometric disturbance to all thermal frames with fixed parameters across the entire video, mimicking practical sensor calibration bias.

Synthetic Misalignment Setting

The synthetic disturbances are designed to reflect typical physical variations in real RGBT deployments.

A horizontal translation can model the lateral baseline discrepancy caused by the left-right mounting distance and calibration tolerance between the RGB and thermal sensors. Similarly, isotropic scaling can approximate the depth-wise mounting discrepancy (front-back displacement) or effective FoV mismatch between two cameras, which leads to a consistent global scale inconsistency. These two factors are thus representative for testing the operating range of alignment robustness in realistic sensing systems. Given a thermal frame I^T , we synthesize a perturbed thermal frame $\tilde{I}^T$ by a 2D warp operator $\mathcal{W}(\cdot)$ that combines translation and isotropic scaling:

$$\tilde{I}^T = \mathcal{W}(I^T; \Delta x, s), \quad (19)$$

where Δx is a horizontal translation (right shift) and s is an isotropic scaling factor around the image center. In coordinate form,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = s \begin{bmatrix} x - c_x \\ y - c_y \end{bmatrix} + \begin{bmatrix} c_x + \Delta x \\ c_y \end{bmatrix}, \quad (20)$$

with (c_x, c_y) being the image center. We use bilinear sampling for warping and keep the original resolution by padding. We evaluate two disturbance: (1) translation $\Delta x \in \{5, 10, 15, 20, 25\}$ pixels, and (2) scaling $s \in \{1.1, 1.2, 1.3, 1.4, 1.5\}$. All models are evaluated without any re-training or fine-tuning on synthetic data.

Results and analysis.

Tab. 9 and Tab. 10 report AP and AP₅₀ under increasing additional synthetic disturbances. To make the turning point more explicit, we additionally report the drop with respect to the no-additional disturbance baseline ($\Delta x = 0$ or $s = 1.0$) in the third row of each table. We empirically treat $\Delta x = 10$ pixels and $s = 1.2$ as the failure thresholds of additional translation and scale, respectively. At these points, the

Table 9: Generalization under synthetic translation disturbances on the UVT-VOD2024 test set. Δx denotes the additional right shift (pixels) applied to thermal frames.

Method	$\Delta x = 0$		$\Delta x = 5$		$\Delta x = 10$		$\Delta x = 15$		$\Delta x = 20$		$\Delta x = 25$	
	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$
IPTNet w/o prog. align.	18.1	36.6	16.6	34.4	14.9	31.8	13.1	28.9	11.7	26.8	10.5	25.0
IPTNet	19.9	41.1	19.2	40.2	18.3	38.7	17.2	36.9	16.0	35.0	14.9	33.4
Drop (w.r.t. $\Delta x = 0$)	0.0	0.0	-0.7	-0.9	-1.6	-2.4	-2.7	-4.2	-3.9	-6.1	-5.0	-7.7

Table 10: Generalization under synthetic scaling disturbances on the UVT-VOD2024 test set. s denotes the isotropic scaling applied to thermal frames.

Method	$s = 1.0$		$s = 1.1$		$s = 1.2$		$s = 1.3$		$s = 1.4$		$s = 1.5$	
	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$	AP $\uparrow$	AP $_{50}$ $\uparrow$
IPTNet w/o prog. align.	18.1	36.6	17.0	35.2	15.6	33.1	14.0	30.7	12.6	28.2	11.0	25.5
IPTNet	19.9	41.1	19.1	40.0	18.0	38.2	16.8	36.0	15.5	33.7	14.1	31.4
Drop (w.r.t. $s = 1.0$)	0.0	0.0	-0.8	-1.1	-1.9	-2.9	-3.1	-5.1	-4.4	-7.4	-5.8	-9.7

average accuracy drop becomes clearly noticeable: AP drops by -1.6 (AP $_{50}$ -2.4) at $\Delta x = 10$, and by -1.9 (AP $_{50}$ -2.9) at $s = 1.2$. Beyond them ($\Delta x > 10$ or $s > 1.2$), the drop accelerates monotonically (e.g., AP drop reaches -5.0 at $\Delta x = 25$ and -5.8 at $s = 1.5$), and qualitative failures become more evident. Our goal is to explore the practical operating range by progressively increasing the additional pixel and scale disturbance until strong degradation is observed. Overall, IPTNet shows a more graceful degradation than the baseline, indicating favorable generalization to unseen extra misalignment. Importantly, even at the extreme settings ($\Delta x = 25$ or $s = 1.5$), a small subset of scenes can still be well aligned (Fig. 10), suggesting a good robustness of the learned transformation. However, the large average drops imply that the majority of scenes already fail under such severe extra misalignment, which is consistent with our qualitative categorization. The scaling disturbances present a consistent trend, further supporting robustness to realistic deployment variations.

4.6 Temporal Metric Evaluation.

While frame-wise AP/mAP remains the standard metric in video object detection, recent studies have proposed temporal-aware variants called VmAP [39] to better reflect performance under temporal association and high-confidence operating points. Following this line, we additionally report VmAP as a supplementary evaluation. Importantly, we do not replace



(a) Raw (b) +25 pixels (c) Aligned

Fig. 10: Visualization of synthetic misalignment construction on UVT-VOD2024 test set. From left to right: raw image overlap, additional translation ($\Delta x = 25$) overlap and aligned image overlap.

the conventional AP/mAP. Instead, VmAP is used to further validate temporal gains under stricter confidence thresholds. We evaluate two settings: (i) single-frame inference ($T=1$), and (ii) multi-frame temporal encoding ($T=2$). We report results under confidence thresholds $\tau \in \{0.9, 0.8\}$, which emphasize high-confidence predictions. As shown in Table 11, incorporating temporal context consistently improves the dataset-level VmAP under both thresholds, indicating

Table 11: Temporal evaluation with VmAP under high-confidence thresholds on UVT-VOD2024. We compare single-frame ($T=1$) and multi-frame temporal encoding ($T=2$).

Temporal Encoding	VmAP $\uparrow$ (%)		Gain vs. $T=1$	
	@ $\tau=0.9$	@ $\tau=0.8$	Δ @0.9	Δ @0.8
Single-frame ($T=1$)	17.60	25.52	+0.00	+0.00
Multi-frame ($T=2$)	24.98	30.95	+7.38	+5.43

Table 12: Per-class VmAP (%) and temporal gain Δ on UVT-VOD2024 under two confidence thresholds. Δ denotes (Multi-frame $T=2$) – (Single-frame $T=1$).

Class	$\tau=0.9$			$\tau=0.8$		
	$T=1$	$T=2$	Δ	$T=1$	$T=2$	Δ
person	40.86	49.10	+8.24	51.80	54.07	+2.27
basketball	3.76	8.63	+4.87	6.19	10.45	+4.26
street-lamp	29.49	35.37	+5.88	36.79	41.42	+4.63
bicycle	0.82	5.67	+4.85	12.57	18.78	+6.21
cat	48.12	64.59	+16.47	59.39	68.32	+8.93
car	10.86	10.27	-0.59	21.15	13.98	-7.17
motorcycle	0.66	8.45	+7.79	6.25	19.89	+13.64
bag	0.57	0.08	-0.49	4.16	4.42	+0.26
bus	2.08	3.24	+1.16	2.66	10.20	+7.54
van	30.94	29.55	-1.39	32.17	32.34	+0.17
football	25.48	59.83	+34.35	47.53	66.63	+19.10

that the temporal encoder enhances reliability in video detection.

Notably, the temporal model yields a clear improvement at $\tau=0.9$ (+7.38 points), suggesting that temporal cues are particularly helpful when we require highly confident detections. Table 12 shows that temporal encoding yields consistent improvements for most categories, especially under the stricter high-confidence threshold $\tau = 0.9$ (e.g., *person*, *cat*, and *football*), indicating that temporal context helps recover difficult instances while maintaining reliable predictions. Notably, categories with fast motion or sporadic visibility (e.g., *football*, *motorcycle*, *bicycle*) benefit substantially from multi-frame cues. We also observe minor drops for a few classes (e.g., *car/van* and *bag* at $\tau = 0.9$), which we attribute to the more conservative behavior introduced by temporal fusion under high-confidence (e.g., stricter suppression/association may reduce recall for small or ambiguous instances). Overall, the per-class results

align with the dataset-level gains in Table 11 and support that our lightweight temporal encoder provides practical benefits for video detection in the alignment-free RGBT setting.

4.7 Visualization of Fusion

To further demonstrate the adaptation of the Thermal-to-RGB fusion module under complex misalignment situations, we visualize the attention weight and the fused feature from shallow layers on the test set. The (b) column of Fig. 8 demonstrates the prominent affinity effects brought by the thermal features after homologous attention, and we can observe that the attention focuses on the foreground within the thermal feature without being influenced by masks in the receptive field. This phenomenon arises from leveraging similarity-based fusion on features extracted from already aligned images, establishing modality complementarity. The fusion of these accurate local similarity makes the model concentrate on the most discriminative features of the two modalities. The foreground within the thermal feature’s receptive field is further enhanced through homologous global modeling. Ultimately, benefiting from the prior alignment, the use of RGB image features as the value, the (c) column clearly displays that the multimodal object features are effectively fused and refined for detection purposes. Additionally, we visualize several scenarios in Fig. 9 from the VT-VOD50 dataset and find that IPTNet assigns higher confidence scores to correctly identified objects. Moreover, it avoids repeated detection for a single object. All the above visualizations demonstrate the effectiveness of the T2RF strategy in IPTNet.

5 Conclusion

The proposed IPTNet model effectively addresses the data misalignment issue in RGBT video object detection by several innovations. First, it achieves end-to-end cross-modal alignment by leveraging a pre-trained ImageBind model, enabling a seamless transition from coarse to fine alignment without the need for additional alignment data or complex manual annotations. Second, IPTNet employs a self-supervised learning approach that utilizes inherent multi-modal prior knowledge to achieve explicit pixel-level alignment, a level of precision that was previously difficult to attain with additional annotation information. Additionally, the T-to-RGB Fusion module focuses on

aligned thermal image information through a homologous attention mechanism, enhancing the efficiency of fusion by better concentrating on the local features of the thermal image. Finally, IPTNet demonstrates strong compatibility and adaptability, effectively handling challenging scenarios such as parallax and resolution differences. These capabilities make the model highly applicable in various real-world scenarios. IPTNet has achieved state-of-the-art performance on UVT-VOD2024 and VT-VOD50, surpassing existing methods and setting a new standard in RGBT VOD, simultaneously achieving optimal performance based on addressing RGBT misalignment. IPTNet excels at handling view and resolution differences but has yet to explicitly address temporal misalignment caused by the time lag between RGB and thermal image, which we plan to tackle it using ImageBind’s video modality in the future.

6 Data Availability Statements

A signed dataset release agreement must be send to qishunahu@163.com to obtain the download link of UVT-VOD2024 and VT-VOD50. CVC-14 is available at <https://pan.baidu.com/s/1sgkiS2ZVv2bMAsUBkjDCSA?pwd=oeji>. LLVIP is available at <https://bupt-ai-cz.github.io/LLVIP/>.

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